Readiness Inspection and Guidance Tool for Joule Scenarios

SAP S/4HANA Cloud Private Edition | FLP Plugin

User Guide & Feature Reference

A practical guide for SAP Admins, Support Engineers, and Functional Experts to verify Joule capability readiness and access on an SAP S/4HANA system.

1. What is Readiness Inspection and Guidance Tool for Joule Scenarios?

Readiness Inspection and Guidance Tool for Joule Scenarios is a plugin for the SAP Fiori Launchpad (FLP). Once deployed on an SAP S/4HANA system, it adds an icon in the FLP Shell bar next to the Joule icon. Clicking that icon opens the Readiness Inspection and Guidance Tool for Joule Scenarios dialog.

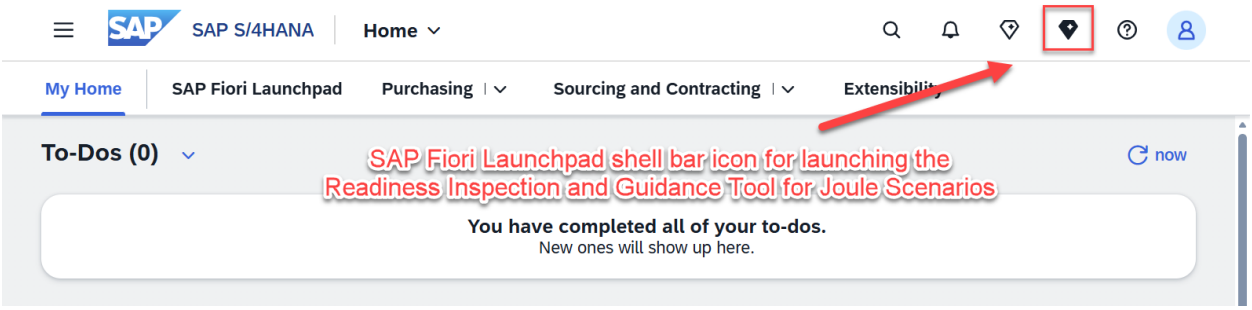


Figure: Readiness Inspection and Guidance Tool for Joule Scenarios icon in the FLP Shell bar

SAP Joule uses a set of capabilities that are deployed on a customer system. Each capability has a set of associated scenarios (1 to N) - also called use cases. Each scenario depends on:

- OData services (APIs) from which it reads data.
- SAP Fiori Apps (IBN Targets, which can be SAPUI5, WebCUIF, Web Dynpro, SAPGUI, or URL type) that it navigates to.

> **Note:** Use cases and scenarios are used interchangeably throughout Readiness Inspection and Guidance Tool for Joule Scenarios and in this document. They refer to the same thing: a specific Joule action within a capability (e.g. "Create Purchase Requisitions" inside "Procurement").

Example — Procurement capability has the following use cases / scenarios:

Use Case / Scenario
Create Purchase Requisitions
Change Purchase Orders
Create Returns Purchase Orders

Readiness Inspection and Guidance Tool for Joule Scenarios provides a statistics and summary view of all capabilities and scenarios present in Joule on the SAP S/4HANA system. It shows how many capabilities are active on the system and how many of them are accessible to the logged on user or a selected context user.

Who is it for?

Persona	How they use it
SAP Admin / Fiori Admin	Understand system readiness; activate missing services and SAP Fiori Apps.
Support Engineer	Diagnose why a specific Joule use case is not working for a user on the customer system.
Functional Expert / SME (Line of Business)	Confirm that a business capability (e.g. Procurement) is fully activated and authorized.
Authorized End User	Check their own capability readiness and access, depending on training

(optional)	level.
Security / Audit Team	Verify how many capabilities are activated on the customer system and authorized to a specific user — to prevent data leaks or misuse.

Key benefits

- Provides readiness stats (activation) and access stats (authorization) for all capabilities and scenarios in one place.
- Lets an Admin switch the whole analysis context to any other user on the customer system - without changing any settings.
- Provides a detailed summary of what needs to be done to fix a specific capability or use case.
- Useful for quality control and security audits - gives tangible visibility of how many capabilities are activated and authorized per user.
- Export's role recommendations directly to Excel.
- Stays current by importing the latest SAP reference data from SAP Note SAP Note 3523238 - Navigational and Transactional capabilities with Joule in SAP S/4HANA Cloud Private Edition or Joule Admin Center. This is the central SAP Note outlining the navigational and transactional capabilities available with Joule in SAP S/4HANA Cloud Private Edition. It also provides a detailed list of SAP Fiori apps enabled in SAP S/4HANA Cloud Private Edition for Joule.

2. Key Concepts

These are the terms used throughout Readiness Inspection and Guidance Tool for Joule Scenarios.

Term	Meaning
Capability	A high-level Joule function deployed on the customer system, e.g. "Purchaser" or "Procurement". Each capability groups one or more scenarios.
Scenario / Use Case	A specific Joule action within a capability, e.g. "Read Purchase Order". Used interchangeably with use case.
OData Service (API)	A backend data service that a scenario reads data from. Must be active on the customer system and authorized for the user.
IBN Target / SAP Fiori App	A Fiori navigation intent required by a scenario. Can be SAPUI5, WebCUIF, Web Dynpro, SAPGUI, or URL type.
Activation / Readiness	Whether the artefact (OData service or SAP Fiori App) is active on the customer system — a system-level readiness check.
Authorization / Access	Whether the logged on user or selected context user has been assigned the required roles to use this artefact.
Health Status	A traffic-light summary combining readiness and access: Green (Healthy), Yellow (Warning), Red (Critical).
Logged On User	The user currently logged into the SAP Fiori Launchpad.
Context User / Simulation	A different user whose readiness and access context the Admin analyses — the whole tool switches to show data for that user.

3. Understanding the Status Indicators

Two independent status indicators are displayed: the Joule Plugin Status, which checks whether the Joule plugin is detected and initialized in the SAP Fiori Launchpad, and the Capability and Scenario Health Status, which evaluates activation and authorization of the required artefacts

3.1 Joule Plugin Status

The plugin status indicator in the FLP Shell bar shows green or red based on whether the Joule plugin is available in the FLP and whether Joule has completely initialized and is ready to be used.

Status	Meaning
Green	The Joule plugin is detected in the SAP Fiori Launchpad and has been initialized successfully.
Red	The Joule plugin is not detected or has failed to initialize.

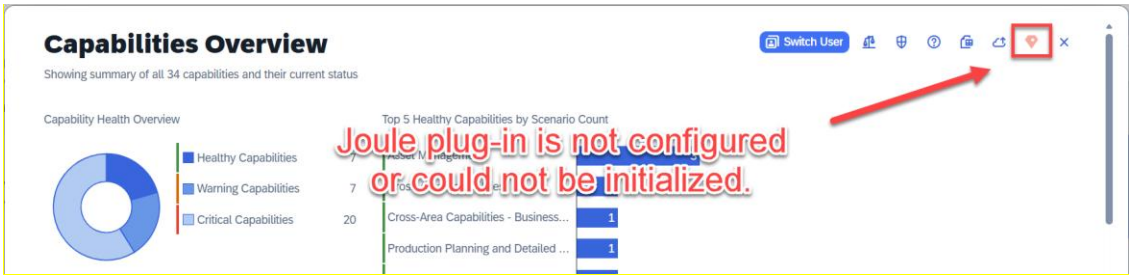


Figure: FLP Shell Bar – Joule plug-in not configured or initialization unsuccessful



Figure: FLP Shell Bar – Joule plug-in successfully initialized and ready for use

3.2 Capability and Scenario Health Status

Every capability, scenario, OData service, and SAP Fiori App is assigned one of three health statuses based on its readiness and access state.

Status	Indicator	What it means
Green — Healthy	**Active + Authorized**	The capability / scenario is fully activated on the customer system (all artefacts are active) AND is fully accessible (the logged-on user or context user is fully authorized).

Yellow — Warning	**Active, not Authorized**	The capability / scenario is ready to be used on the customer system from a readiness point of view (fully activated), but the user may not have access to at least one scenario artefact (at least one OData service or SAP Fiori App is missing authorization).
Red — Critical	**Not Active**	At least one scenario artefact (OData service or SAP Fiori App) is inactive on the customer system. The user may also be missing authorization.

Note: A capability turns Critical as soon as any one of its scenarios is Critical. Fix the Critical scenarios first, then address Warning ones.



Figure: Health Status

4. What Can Readiness Inspection and Guidance Tool for Joule Scenarios Do?

Readiness Inspection and Guidance Tool for Joule Scenarios is organized into 12 screens, all inside a single dialog launched from the FLP Shell bar.

#	Screen	How to open it	What it does
1	**Capabilities Overview**	Opens automatically on launch	Statistics and summary of all capabilities and their health on the customer system.
2	**Switch User / Context User Simulation**	***Capabilities Overview*** screen only — opens user selection dialog. On other pages the button is disabled and shows the context username as	Switch the whole analysis context to any other user on the customer system.

		a status label.	
3	**Scenarios Overview**	Click any row on the Capabilities Overview screen	Statistics and summary of all use cases inside the selected capability.
4	**Individual Scenario Detail**	Click any row on the Scenarios Overview screen	Full readiness and access breakdown for one use case — APIs and SAP Fiori Apps.
5	**Target Mapping Role Analyzer (TMA)**	Click the ***Analyze missing IBN Targets*** button or the inspect icon next to an unauthorized IBN target on the Scenario Detail page.	Find which business roles are needed to authorize a specific Target Mapping.
6	**API Authorization Analyzer (APIA)**	Click the ***Analyze missing*** **OData Auth** button or the inspect icon next to an unauthorized API on the Scenario Detail page	Find which business roles are needed to authorize a specific OData service.
7	**Activation Summary (only for SAPUI5 type IBN Targets)**	Detail button on an Activating Warning for a SAP Fiori App (IBN Targets) row	Show the dependency chain behind why an SAP Fiori App is only partially active.
8	**Resolve Errors**	***Resolve Errors*** button on the Scenario Detail screen	Step-by-step guide to fix every activation and authorization issue in the current scenario.
9	**Prompt Library**	Help icon button in the screen header	Browse Joule use case prompts from the SAP Help documentation and send them directly to Joule.
10	**Capability Data Sync**	Excel icon button in the Capabilities Overview header	Upload or export the capability and scenario reference data from SAP Note 3523238.
11	**Custom Role Analyzer (CRA)**	Shield icon button in the Capabilities Overview screen header	Analyse authorization coverage of one or more business roles across all Joule scenarios.
12	**Import Joule Admin Center Data**	Button in the Capabilities Overview screen header	Upload Joule Admin Center scenario visibility data to sync all scenarios and their required IBN Target or intent-based navigation entries.

5. Screen-by-Screen Guide

5.1 Capabilities Overview

After launching Readiness Inspection and Guidance Tool for Joule Scenarios from the FLP Shell bar, the user sees the Capabilities Overview screen. This screen provides statistics and summary of all capabilities present in Joule on the customer system and their current health for the logged on user or selected context user.

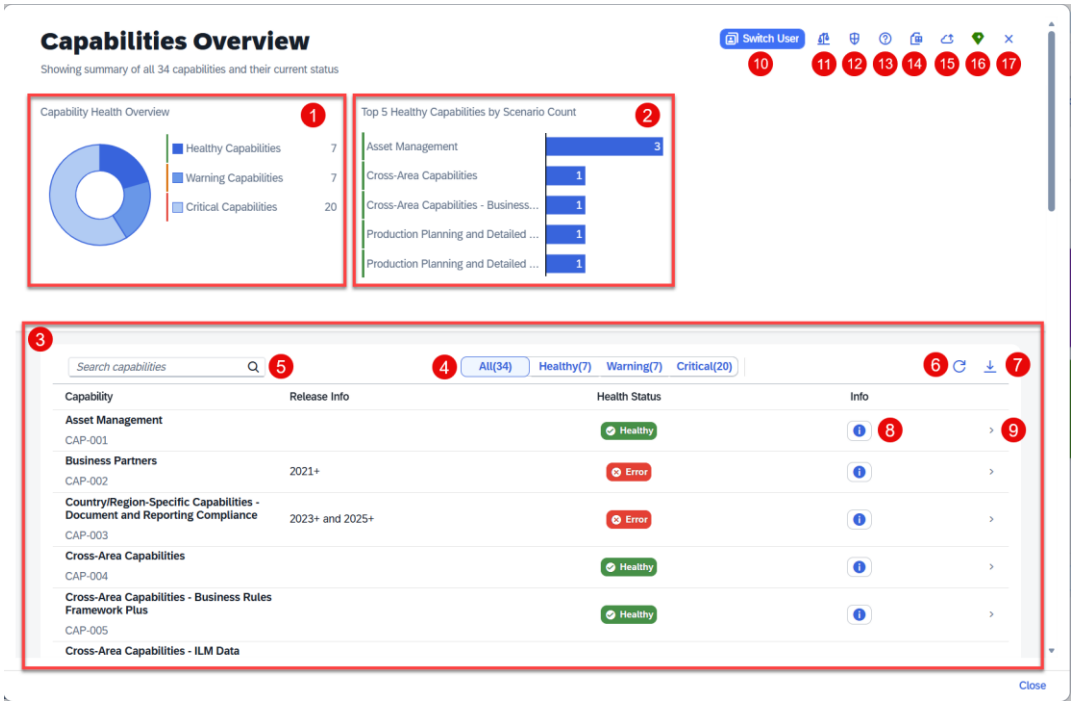


Figure: Capabilities Overview screen

Element	Description
Capability Health Overview - Donut chart (1)	Stats about readiness and access to capabilities as a whole for the logged on user or selected context user. Shows counts of Healthy, Warning, and Critical capabilities.
Top 5 Healthy Capabilities - Bar chart (2)	Counts the top 5 healthy capabilities by scenario / use case count — useful for understanding which capabilities with multiple use cases are fully accessible.
Capabilities table (3)	Full list of all capabilities with their name, release info, and health status. Clicking any row opens the Scenarios Overview for that capability.
All / Healthy / Warning / Critical filter (4)	Narrows the capabilities table to the selected health status.
Search (5)	Filter the capabilities table by capability name, release info, or health status.
Refresh (6)	Reload all capability data from the customer system.
Download (7)	Download the current capabilities list to an Excel file (Capabilities_Overview.xlsx). Includes capability name, ID, release info, and health status.
Info (ⓘ) (8)	Opens the Capability Detail screen for the selected capability. Shows a full breakdown of OData service and SAP Fiori App activation and authorization status for that capability.
Open Scenarios (row press) (9)	Click any row in the capabilities table to open the Scenarios Overview, showing all use cases (scenarios) within that capability and their readiness status.
Switch User (10)	Switch the whole analysis context to another user on the SAP S/4HANA system.

Compare Roles (11)	Open the Role Federation Comparison screen to compare the selected user's role assignments across SAP S/4HANA, Common Data Model (CDM) V1 and V2.
Custom Role Analyzer (CRA) (12)	Open the Custom Role Analyzer to check authorization coverage of one or more business roles across all Joule scenarios in the system.
Prompt Library (13)	Open the Joule Prompt Library.
DB Sync (14)	Open the Capability Data Sync screen.
Import Admin Center Data (15)	Open the Import Joule Admin Center Data screen to upload scenario visibility data exported from the SAP Joule Admin Center.
Joule Plugin Status (16)	Visual status indicator for the SAP Joule plugin, as described in Section 3.1.
Close (17)	Close the tool and return to the SAP Fiori Launchpad.

5.2 Switch User — Context User Simulation

The ***Switch User*** button is available on the ***Capabilities Overview*** screen (main page). On all other pages the button is shown as a disabled status label, and it displays the name of the context user currently being simulated (e.g. "***Simulating THOMASE***") but does not open the selection dialog. To switch to a different user, return to the ***Capabilities Overview*** first.

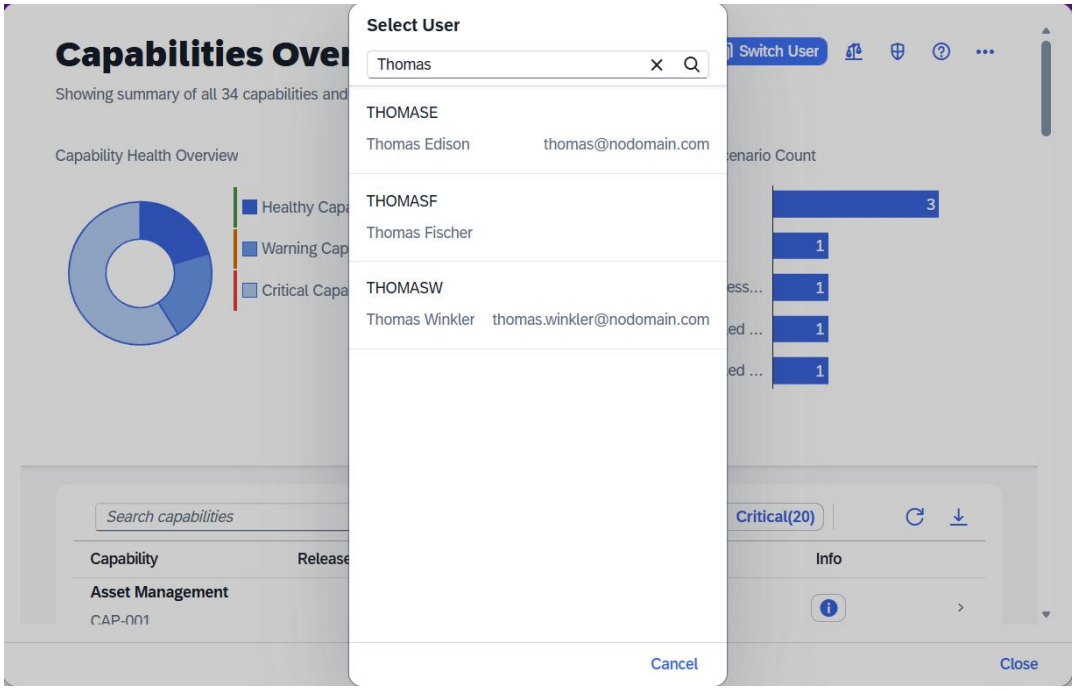


Figure: Switch User — simulation mode active for a context user

- Press ***Switch User*** to open the user selection dialog. Search for and select any user on the system.
- A yellow Message Strip banner appears on the ***Capabilities Overview*** screen as a reminder that simulation is active.

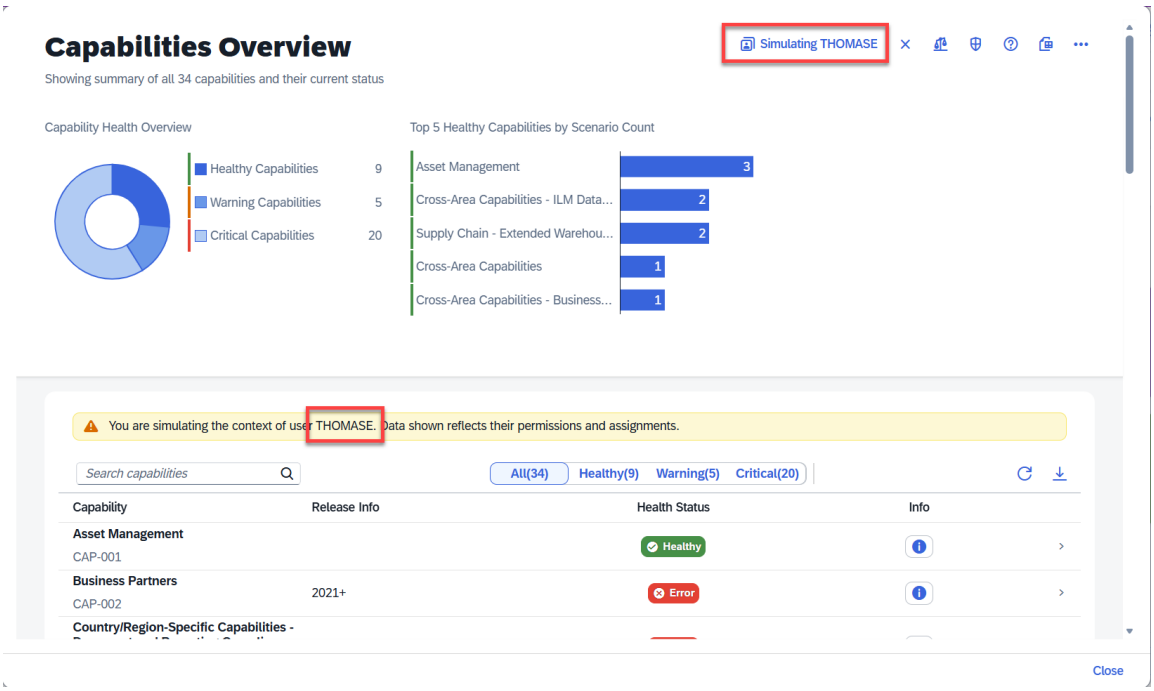


Figure: Switch User — simulation mode active for a context user

- The ***Switch User*** button label changes to show the name of the selected context user.
- All data i.e. capabilities, scenarios, OData services, SAP Fiori Apps - is now loaded for the context user.
- Press Cancel Simulation to revert to the logged-on user. Capability data reloads automatically.

Note: Context user simulation is read-only. No settings on the customer system are changed - the tool only reads data on behalf of the selected user.

5.3 Scenarios Overview

Clicking any row on the ***Capabilities Overview*** screen opens the ***Scenarios Overview***. This screen shows all scenarios (use cases) inside the selected capability.

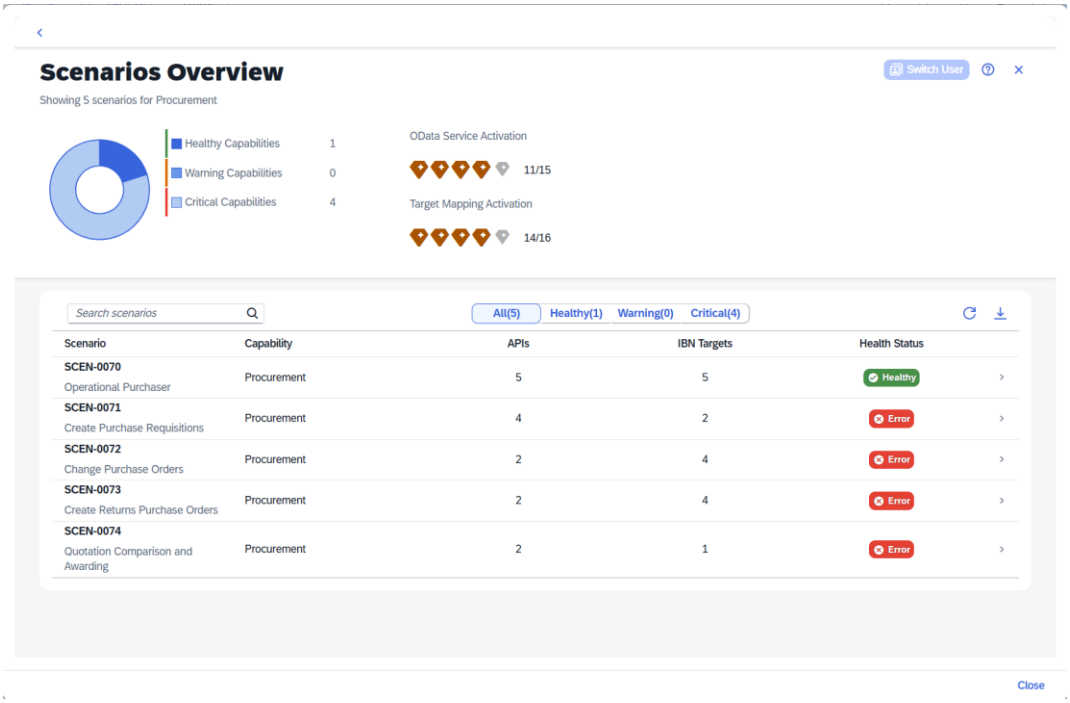


Figure: Scenarios Overview screen

- Scenarios Health Overview: Stats about readiness and access to use cases inside the capability for the logged on user or selected context user.
- The traffic light applies the same Green / Yellow / Red logic as the Capabilities Overview but at the scenario level.
- A summary shows how many OData services (APIs) and SAP Fiori Apps (IBN Targets) are active in that capability.
- The scenario list shows each use case name, description, namespace, target function, and health status.
- Clicking a scenario row opens the Individual Scenario Detail screen.
- This screen also works based on the logged-on user or context user selected on the Capabilities Overview screen.

5.4 Individual Scenario Detail

This screen highlights a specific use case (scenario) and shows the status of its artefacts i.e. the APIs (OData services) and SAP Fiori Apps (IBN Targets) it depends on.

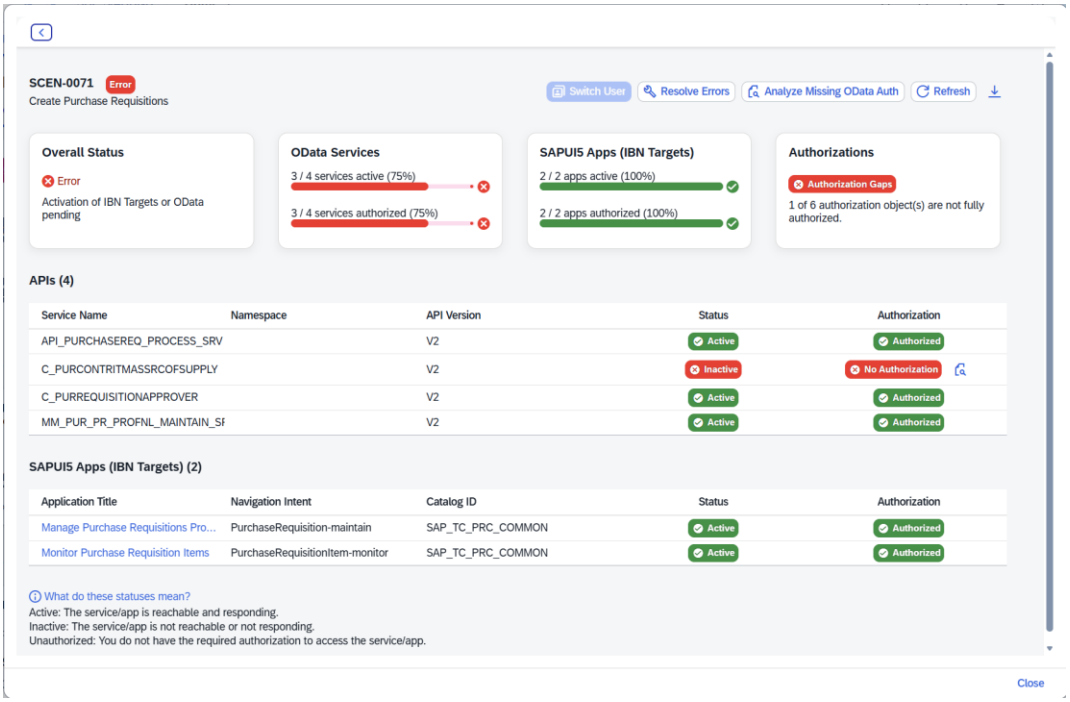


Figure: Individual Scenario Detail — summary cards, APIs, and SAPUI5 Apps (IBN Targets)

Summary cards

Card	What it shows
Overall Status	**Red / Error**: at least one TM or OData service is inactive on the customer system. **Yellow / Warning**: all TMs and OData services are active on the customer system, but at least one is not authorized for the user. **Green / Success**: all TMs and OData services are active AND the logged on user or context user has access to all of them.
APIs (OData Services)	Count of OData services required by this use case: how many are active and how many are authorized for the user.
SAPUI5 Apps (IBN Targets)	Count of SAP Fiori Apps (IBN Targets) required by this use case: how many are active and how many are authorized for the user.
Authorizations	Overall authorization summary for this use case.

APIs (OData Services) section

Shows the activation status and authorization status of each OData service required by this use case, along with namespace and API version.

Column	Description
Service Name	Technical name of the OData service.
Namespace	Service namespace.
API Version	V2 or V4.
Status	Activation status: Active, Warning, or Inactive on the customer system.
Authorization	Authorization status for the logged-on user or context user: Authorized or No

	Authorization. An Inspect button appears on No Authorization rows to launch APIA.

SAP Fiori Apps (IBN Targets) section

Shows the activation status and authorization status of each SAP Fiori App (IBN Target) required by this use case.

Column	Description
Application Title	App name, linked to the Fiori Apps Library.
Semantic Object	Navigation semantic object.
Action	Navigation action.
Status	Activation status on the customer system: Active, Warning, or Inactive. A detail button on Warning rows opens the Activation Summary.
Authorization	Authorization status for the logged-on user or context user: Authorized or No Authorization. An Inspect button appears on No Authorization rows to launch TMA.

Action buttons on this screen

Button	Appears when	What it does
Analyze Missing IBN Targets	At least one SAP Fiori App (IBN Target) has No Authorization	Launches TMA for all unauthorized SAP Fiori Apps (IBN Targets) at once.
Analyze Missing OData Auth	At least one OData service has No Authorization	Launches APIA for all unauthorized OData services at once.
Resolve Errors	Any activation or authorization issue exists	Opens the Troubleshooting Help screen with a step-by-step action plan.
Refresh	Always	Reloads all artefact data for this use case.
Export	Always	Downloads the scenario detail data.

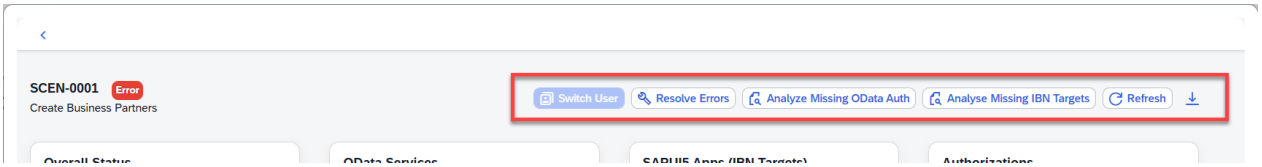


Figure: Action buttons on Individual scenario details screen

5.5 Target Mapping Role Analyzer (TMA)

Here you can analyze all the IBN Targets (SAP Fiori Apps) that have the required Intent (Semantic Object and Action) and identify which business roles and composite roles need to be assigned to the user. Work left to right through the three columns.

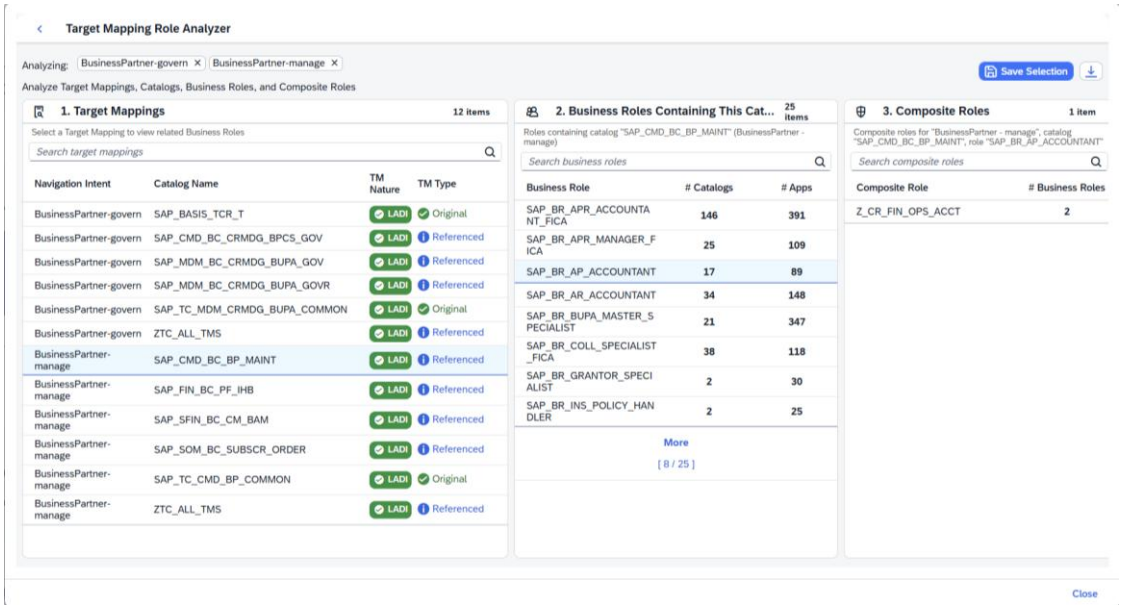


Figure: Target Mapping Role Analyzer (TMA) – 3 column view

Column	Content	What to do
1. Target Mappings	All TMs matching the selected Semantic Object and Action. Shows navigation intent, catalog name, catalog type (LADI / TM), and TM type (Original / Referenced).	Pick a TM + Catalog as input to load Business Roles in Column 2.
2. Business Roles	Business roles containing the catalog of the selected TM. Shows role name, catalog count, and app count.	Pick a business role containing the TM and catalog from Column 1 to load Composite Roles in Column 3.
3. Composite Roles	Composite roles that contain the selected business role.	Pick the composite role. Then press Save Selection to record the full chain.

- A Tokenizer at the top shows the active **Semantic Object–Action** pairs being analyzed. Individual tokens can be removed to narrow scope.
- Search is available in each column.
- ***Save Selection*** saves the current TM + Business Role + Composite Role combination. The button shows the running saved count.
- ***Download*** exports all saved selections to an Excel file (Required_Roles_for_TMs_Export.xlsx).
- Saved TMA selections appear as recommended role assignments for the corresponding IBN Targets in ***Resolve Errors***. See the figure **"Resolve Errors — Saved Role Recommendations from TMA and APIA"** in Section 5.8.

5.6 API Authorization Analyzer (APIA)

Here you can analyze all the OData APIs and identify which business roles need to be assigned to the user. Use the same three-column approach as TMA, but for OData services.

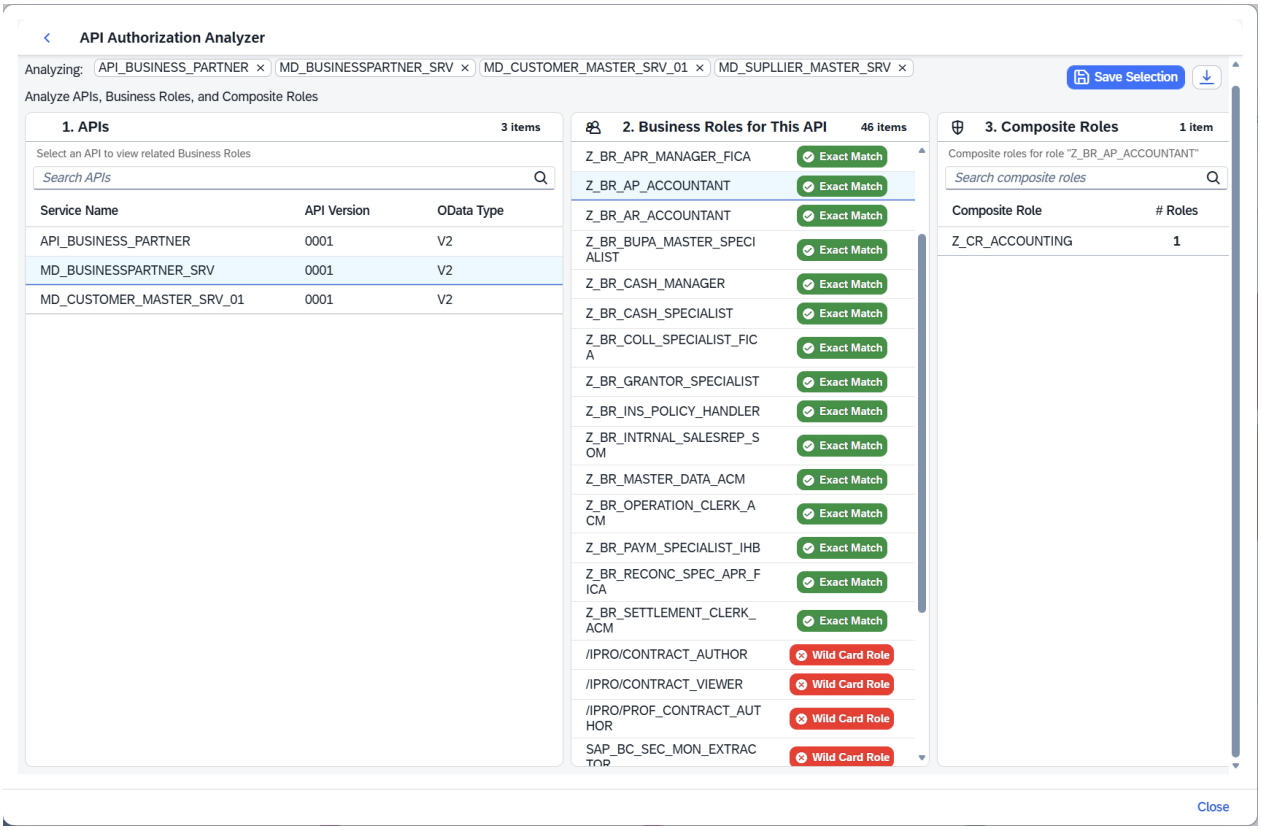


Figure: API Authorization Analyzer (APIA) — 3-column view

Column	Content	What to do
1. APIs	OData services being analyzed (name, version, OData type).	Pick an OData service to load its Business Roles in Column 2.
2. Business Roles	Roles that contain the authorization for the selected OData service, with Authorization Match Type.	Pick a business role to load Composite Roles in Column 3.
3. Composite Roles	Composite roles containing the selected business role.	Pick the composite role, then press Save Selection.

Authorization Match Type for roles in Column 2:

Match Type	What it means	Recommendation
Exact Match	Role has exact authorizations for this specific OData service.	**Recommended** - most precise option.
IWSG Auth only	Role authorizes SAP Gateway: Service Groups Metadata	**Use with caution** - partial authorization.
IWSV Auth only	Role authorizes SAP Gateway Business Suite Enablement – Service.	**Use with caution** - partial authorization.
Wild Card	Role uses a wildcard authorization covering many OData services.	**Not recommended** - overly broad, security risk.

Note: Always prefer Exact Match roles. Wildcard roles grant access beyond what is intended for this use case.

- **Exact Match** roles contain the specific S_SERVICE authorization object required for this OData service — this is the most precise and safest assignment.
- **Wildcard** roles use a * in the S_SERVICE authorization fields, which grants access to all OData services at once. They work in practice but are not recommended for production because they grant broader access than any single scenario requires.
- **IWSG Auth only** roles contain partial authorization. Verify that the corresponding IWSV authorization is available through another assigned role. If missing, add it to ensure complete service access.
- **IWSV Auth only** roles contain partial authorization. Verify that the corresponding IWSG authorization is available through another assigned role. If missing, add it to ensure complete service access.

Use **Save Selection** to record the selected Business Role and Composite Role for each OData service. These selections appear as recommended role assignments in **Resolve Errors**. See the figure **"Resolve Errors — Saved Role Recommendations from TMA and APIA"** in Section 5.8.

5.7 Activation Summary

Opened from the detail button on a Warning SAP Fiori App (IBN Target) row in the Scenario Detail screen. Shows the full dependency chain behind why the SAP Fiori App is only partially active on the customer system.

Panel	What it shows
App Info	SAP Fiori App name, Fiori App ID, Semantic Object and Action, Catalog, and a link to the Fiori Apps Library.
SAPUI5 Component / Library Dependencies	Frontend UI5 library components the app requires. Activation status and status text per component.
OData V2 Dependencies	OData V2 services this SAP Fiori App depends on. Activation status and reason text per service.
OData V4 Dependencies	OData V4 services this SAP Fiori App depends on. Activation status and reason text per service.

Note: Use this screen to identify the root cause of a Warning status. Typically, one dependent OData service or library needs to be activated using a Rapid Activation task list (transactions STC01 / STC02) on the customer system.

5.8 Resolve Errors

Press the ***Resolve Errors*** button on the Scenario Detail screen. Readiness Inspection and Guidance Tool for Joule Scenarios automatically builds a **step-by-step action plan** to solve every activation and authorization issue found in the current use case.

For the most targeted guidance, use ***Analyze Missing OData Auth*** and ***Analyze Missing IBN Targets*** before opening this screen. In each of those screens, use ***Save Selection*** to record the Business Role and Composite Role combination identified for each failing service or app.
When you then open ***Resolve Errors***, those saved selections appear inline inside the relevant issue card as direct role recommendations - turning the general remediation steps into a specific action list for your Basis team.

Issue found	Guidance given
OData service is inactive on the customer system	Steps to activate it using the SAP Rapid Activation task list (STC01 / STC02).
OData service — user has No Authorization	Steps to identify the required business role using APIA, set it up in PFCG, and assign it to the user.
SAP Fiori App (IBN Target) is inactive on the customer system	Steps to activate Fiori content and ICF nodes using STC01 / STC02.
SAP Fiori App has inactive dependencies	Steps to activate all content for the affected catalog or business role.
SAP Fiori App (IBN Target) — user has No Authorization	Steps to assign the required PFCG role to the user directly or via a composite role.

- Each issue card shows a category label (Activation or Authorization), a title, numbered action steps, and an action button that navigates directly to the relevant analyzer.
- If you save role selections in TMA or APIA, they appear inline in the relevant card as recommended role assignments.
- Copy to Clipboard puts the full guide in plain text for sharing via a support ticket or chat.
- Send via Email opens the default mail client with a pre-filled subject and body — addressed to the logged-on user or to an Admin when in context user simulation mode.

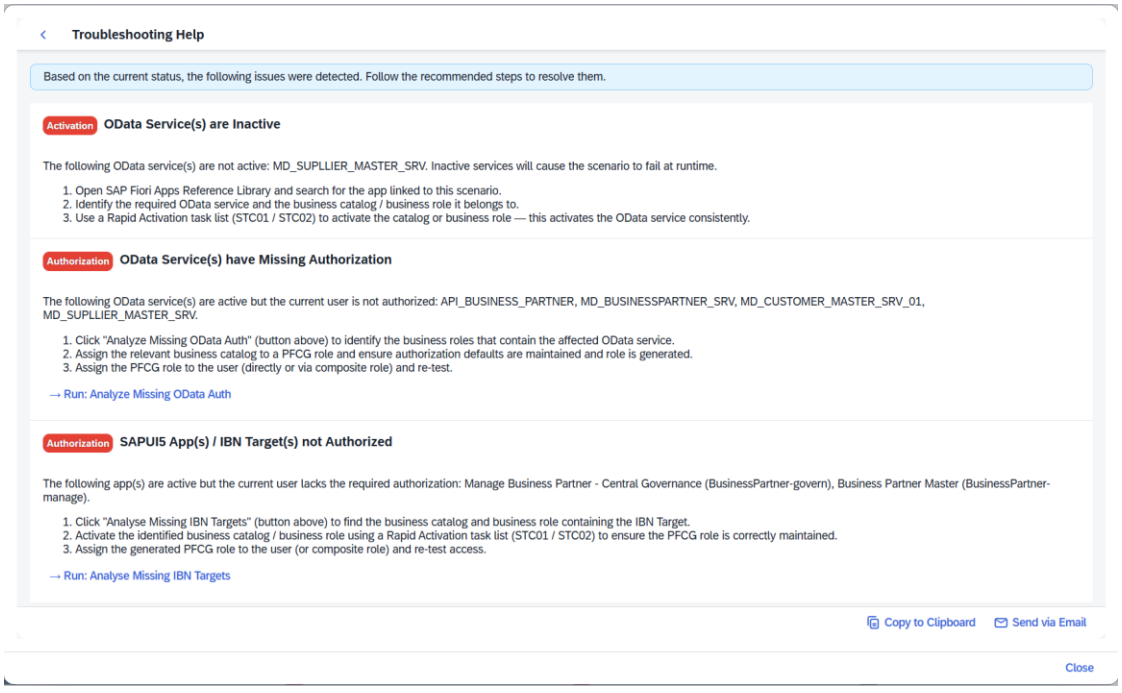


Figure: Resolve Errors — Step-by-Step Activation and Authorization Guidance

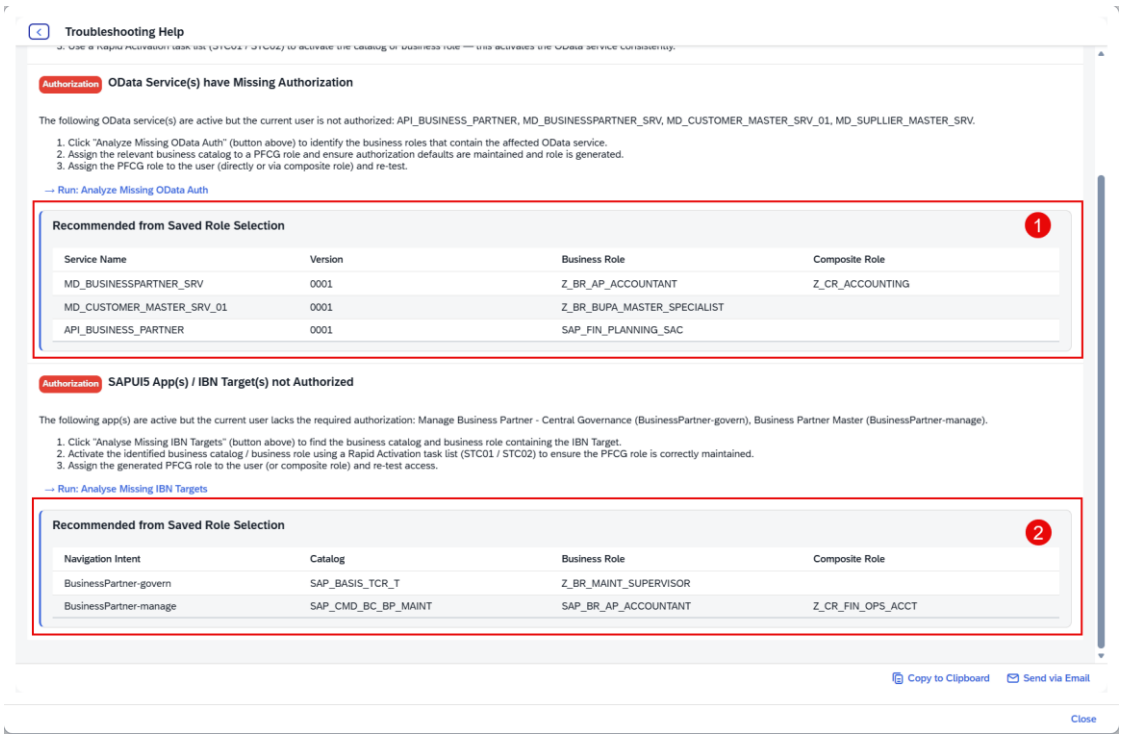


Figure: Resolve Errors — Saved Role Recommendations from APIA (1) and TMA (2)

5.9 Joule Prompt Library



Figure: Joule Prompt Library button (Help Icon) in the Capabilities Overview screen header

This library contains various use case prompts published by the SAP Help documentation listing Joule capabilities. Prompts can be sent directly from the tool to Joule or via editing the text first. Open it from the help icon button in any screen header.

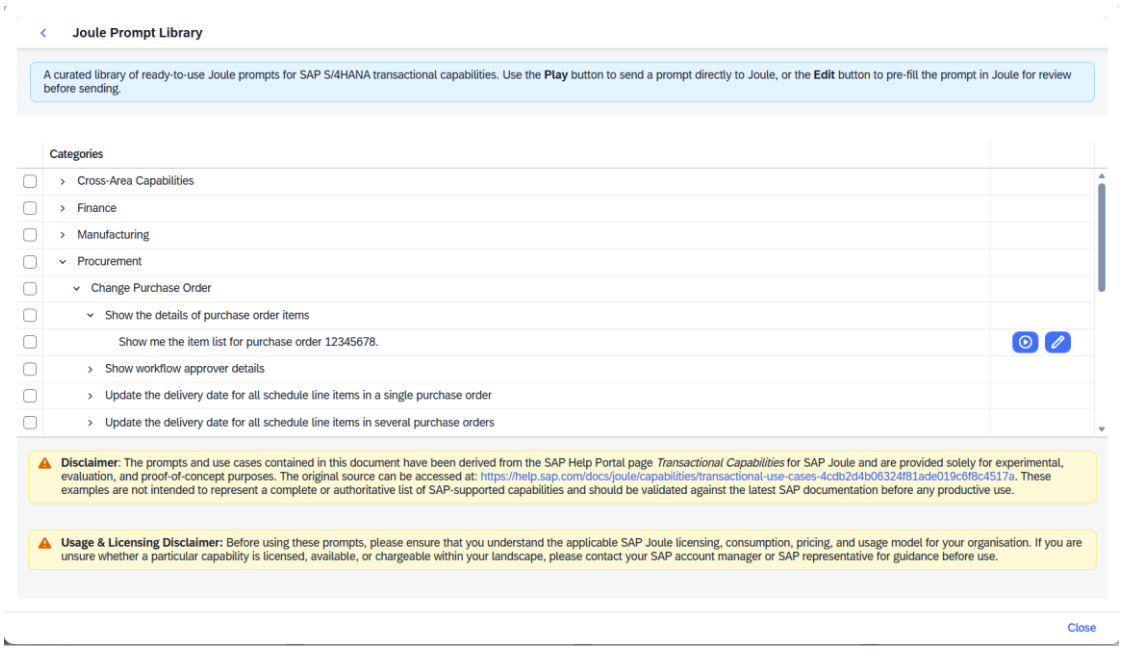


Figure: Joule Prompt Library

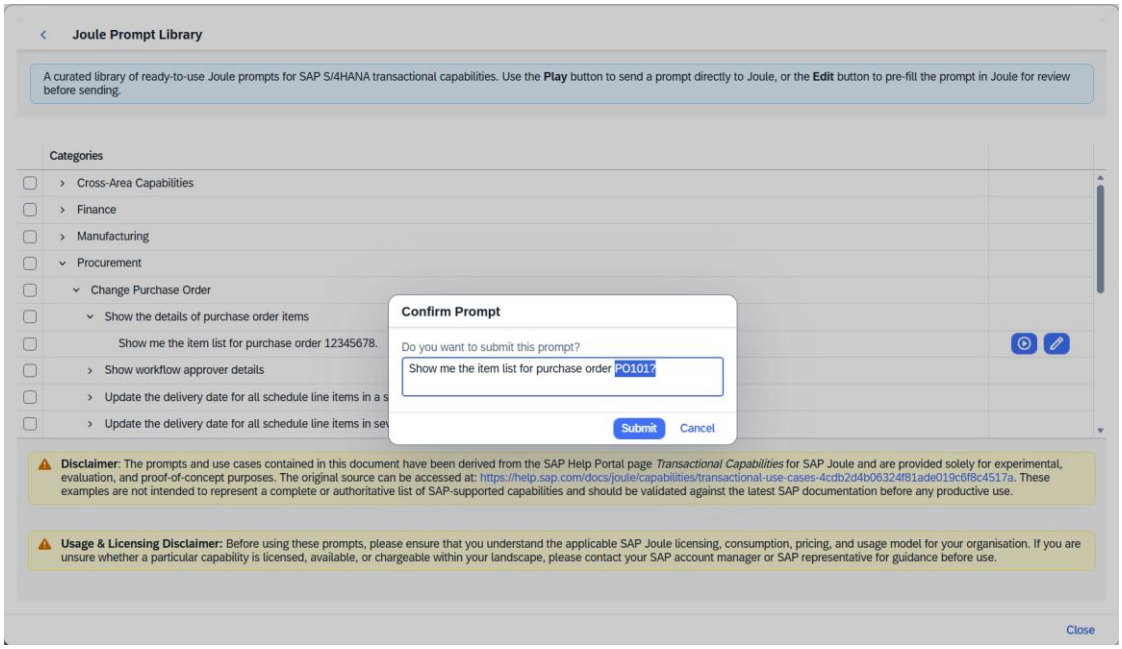


Figure: Prompt Library — tree view of use cases and prompts

- Prompts are displayed in a tree table organized by capability category. Expand a category to see individual use case prompts.
- Play button — sends the prompt directly to the Joule assistant without modification.
- Edit button — opens the prompt in a text box for review and editing before sending to Joule.

Note: Always verify your organization's SAP Joule licensing and consumption model before using prompts in a productive environment.

5.10 Capability Data Sync



Figure: Capability Data Sync button (Excel Icon) in the Capabilities Overview screen header

The mapping of capability use cases (scenarios) to IBN Targets and APIs is provided for customers via SAP Note 3523238. This information can be imported into Readiness Inspection and Guidance Tool for Joule Scenarios where it persisted as the tool's data source. This screen ***Capability Data Sync*** allows to upload the latest version of this data or export it.

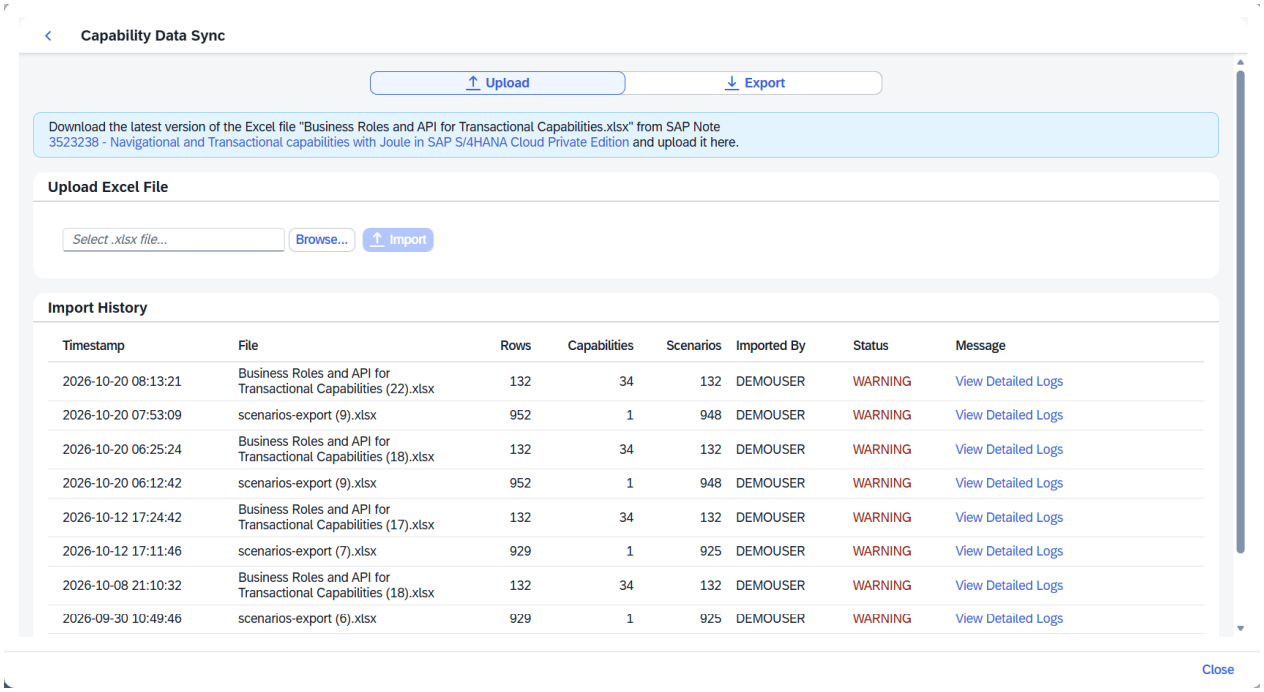


Figure: Capability Data Sync — upload mode

Mode	How to use
Upload	Download the latest version of the Excel file "Business Roles and API for Transactional Capabilities.xlsx" from SAP Note 3523238. Select the file here (max 10 MB) and press Import. A status strip shows the progress.
Export	Download the current capability and scenario data stored on the customer system as an Excel file. Use as a backup or as a template for custom capabilities.

The ***Capability Data Sync*** screen shows who updated the data from this note and persists only the latest data in tables based on the uploaded Excel. The Import History table at the bottom shows the last 20 imports — use the ***View Logs*** link to validate each import:

Column	Content

Timestamp	When the import ran.
File	The uploaded filename.
Rows	Number of rows in the file.
Capabilities	Capabilities created or updated.
Scenarios	Scenarios created or updated.
Imported By	The user who ran the import.
Status	Success, Warning or Error.
View Logs	Open the detailed import log for each sync run. The log shows which rows were processed successfully and which could not be imported — for example due to unrecognized values, typos in the Excel data, or content not yet supported in the current release.

5.11 Custom Role Analyzer (CRA)

The ***Custom Role Analyzer*** (CRA) allows to verify whether one or more business roles provide sufficient authorization coverage for Joule scenarios. Open the CRA by pressing the shield icon button (third button in the Capabilities Overview screen header).

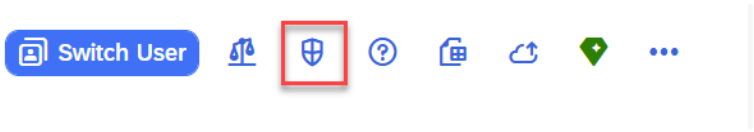


Figure: Custom Role Analyzer button (shield icon) in the Capabilities Overview screen header

The CRA home screen shows the role input area, analysis mode selection, and the Analysis History table for previous analyses.

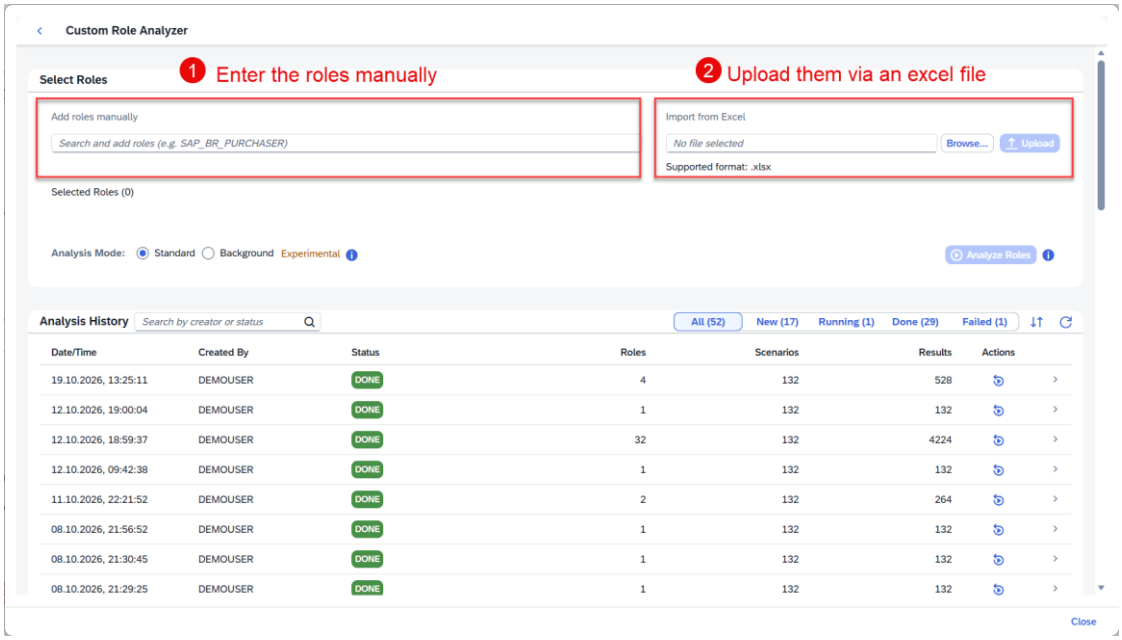


Figure: Custom Role Analyzer home screen

Roles can be added in two ways:

- **By name (type-ahead)**: Start typing a business role name in the input field. Live type-ahead suggestions appear from the business roles available on the system. Select a suggestion to add the role to the list.

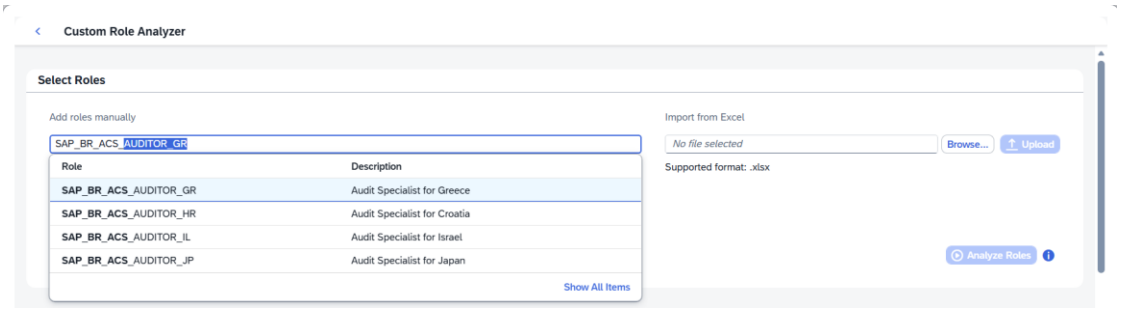


Figure: Adding a role by name with type-ahead support

- **From an Excel file**: Upload an Excel file containing the role names. Any filename is accepted. The role names must appear **in the first column**. All roles found in the file are imported into the list.

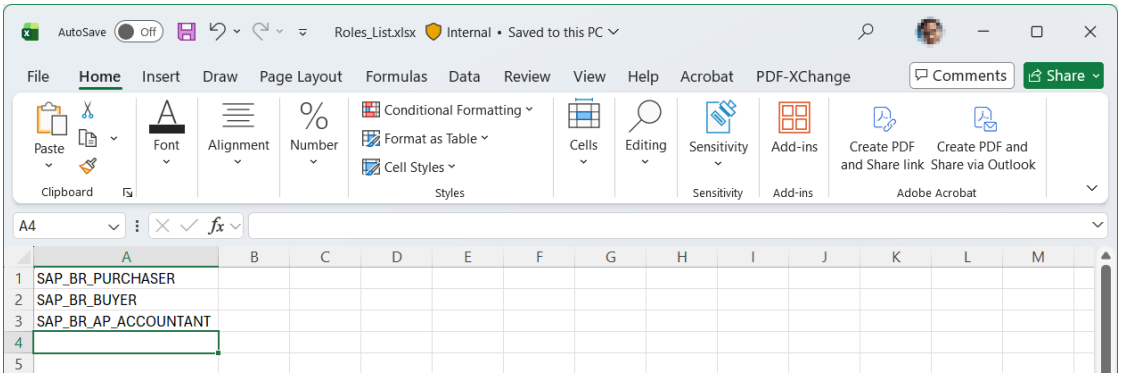


Figure: Role added to the input list

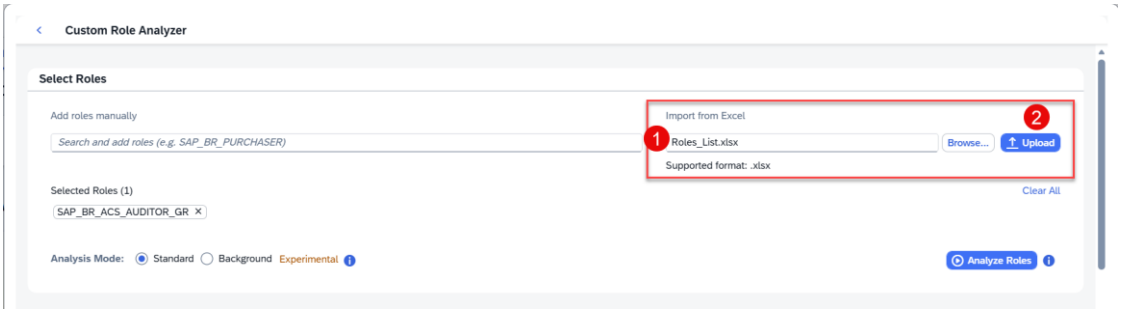


Figure: Excel file upload for role import

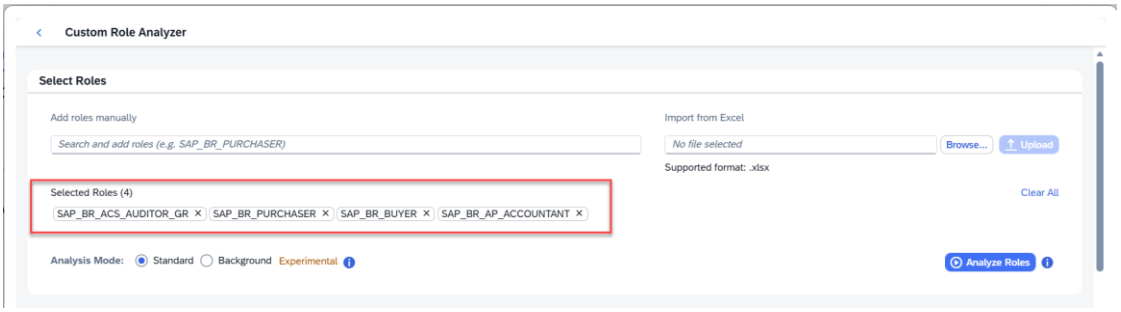


Figure: Unique combined role list ready for analysis

Both methods can be combined. The result is a unique, deduplicated list of roles ready for analysis.

Select the analysis mode and press Analyze:

- **Standard mode**: Runs the analysis immediately in a dialog. Suitable for small to medium role sets.
- **Background mode**: Runs the analysis as a background job. Recommended for large role sets. Requires SAP S/4HANA 2025 with the background processing framework (bgPF and bgRFC) configured and the BGMC_DEFAULT background processing context active.

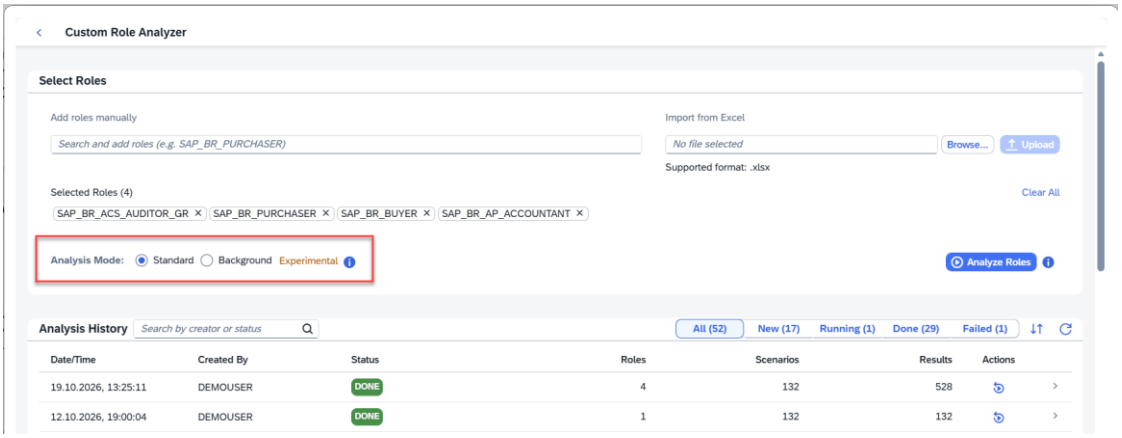


Figure: Selecting Standard or Background analysis mode

Note: Background Analysis is currently **experimental** and has only been tested and validated on **SAP S/4HANA 2025**. It requires the ABAP Background Processing Framework (bgPF), bgRFC configuration, an active BGPF inbound destination, and the default background processing context (**BGMC_DEFAULT**). If these prerequisites are not met, use Standard mode.

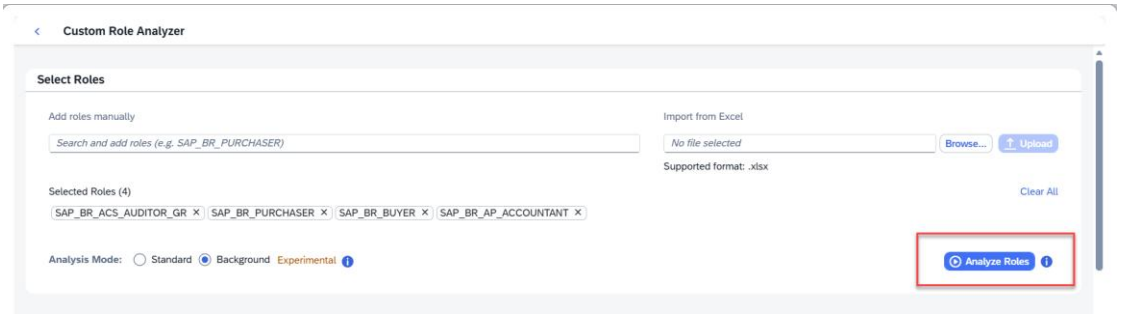


Figure: Analysis configuration ready — press Analyze to start

Note: The CRA checks basic authorization coverage for Target Mappings and OData APIs only. It does not validate functional, business, organizational, application specific, or data level authorizations. This is not a complete functional authorization check.

After the analysis completes, it appears in the ***Analysis History*** table. The table shows all previous analyses with their status and the user who ran them. Use the search and filter controls to locate a specific past analysis or re-run a previous one as a new analysis.

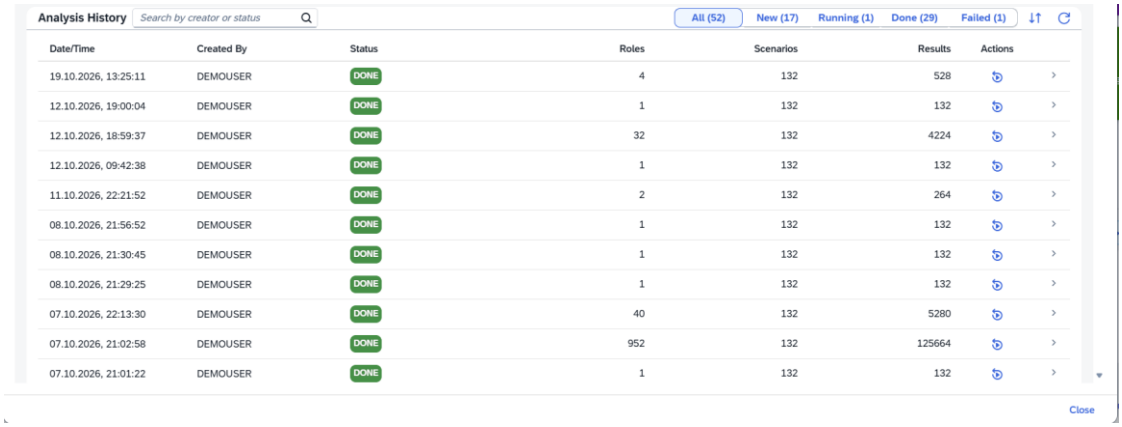


Figure: Analysis History table showing past analyses

Click any row in the Analysis History table to open the analysis results. The results screen shows a coverage matrix: for N selected roles and M scenarios in the system, the matrix has up to N × M cells.

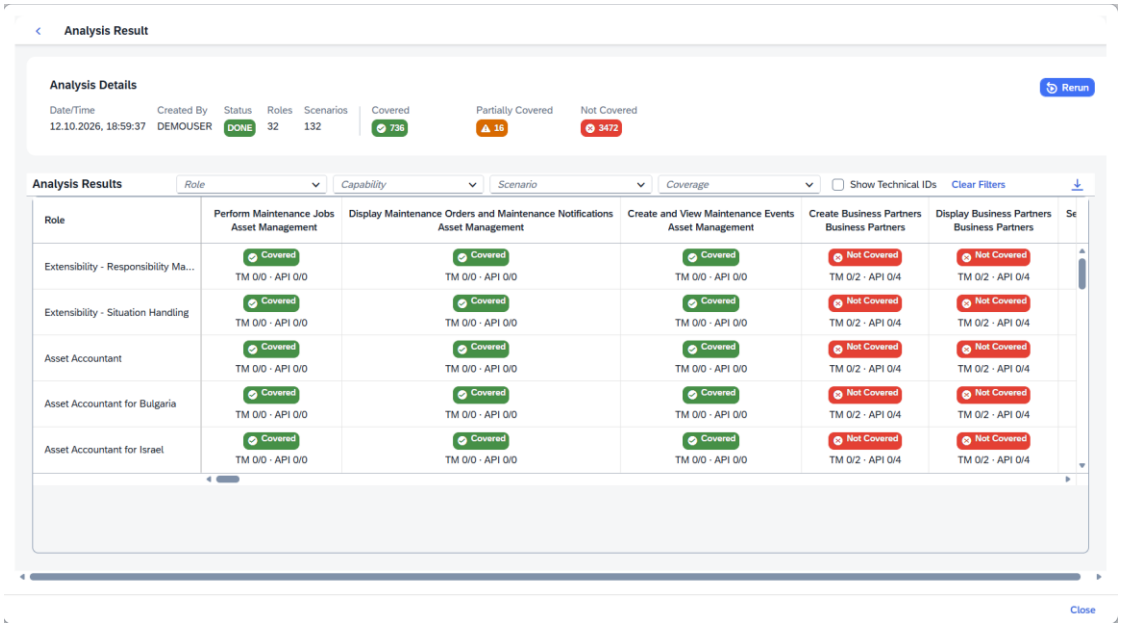


Figure: Analysis results coverage matrix

- **Covered**: The role provides all required authorizations for this scenario.
- **Partially covered**: The role provides some but not all required authorizations for this scenario.
- **Not Covered**: The role does not provide the required authorizations for this scenario.

Click any cell in the matrix to view the missing requirements and overall rating for that specific role and scenario combination.

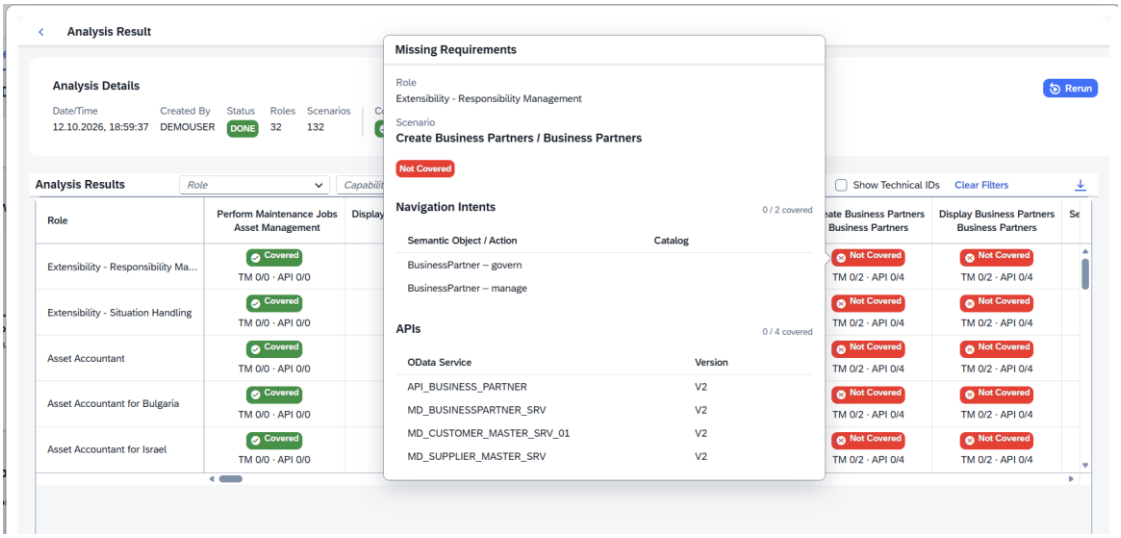


Figure: Missing requirements detail for a selected cell in the coverage matrix

Results can be filtered by Role, Capability, Scenario, and Coverage status.

5.12 Import Joule Admin Center Data (Excel)

The ***Import Joule Admin Center Data*** screen allows to upload scenario visibility data that has been exported from the **SAP Joule Admin Center**. This data describes which applications and prerequisites are required for each Joule scenario.



Figure: Import Joule Admin Center Data button in the Capabilities Overview screen header

On this screen you can upload an Excel file containing the Joule Admin Center scenario visibility data.

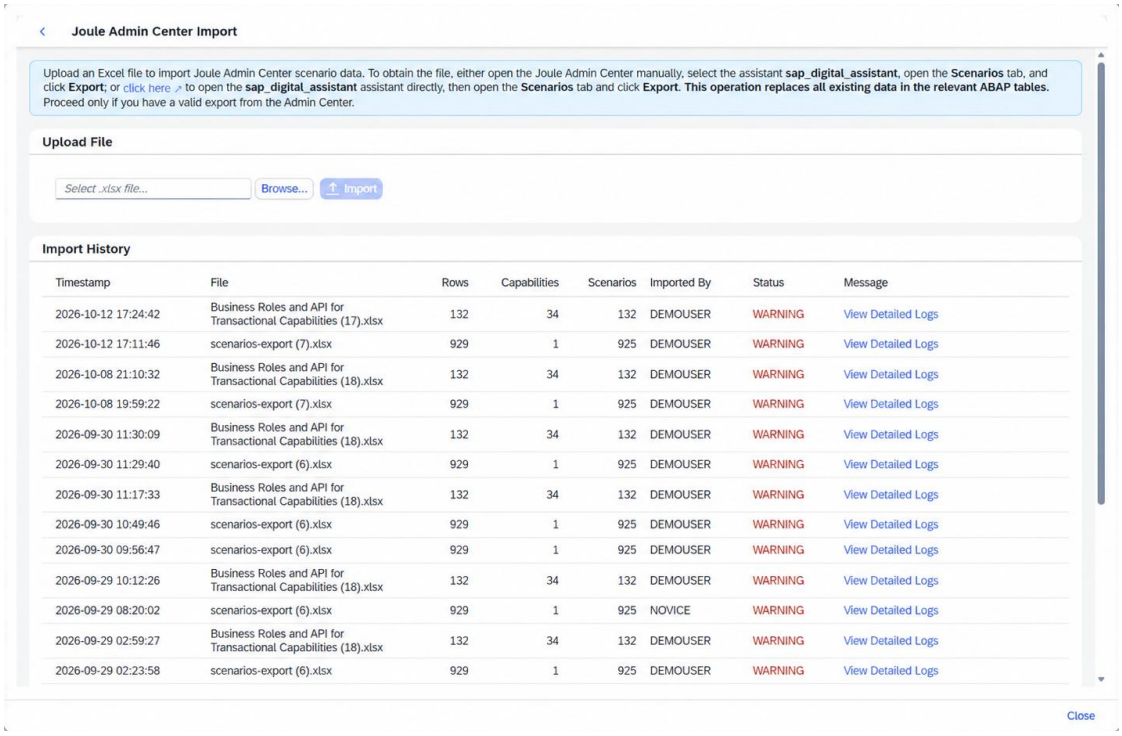


Figure: Import Joule Admin Center Data — upload screen

The import syncs the database with all the scenarios and their required IBN Target or intent-based navigation entries. After a successful import, a single capability named "***Joule Admin Center – Data***" is created in the ***Capabilities Overview***. All imported scenarios are grouped under this capability.

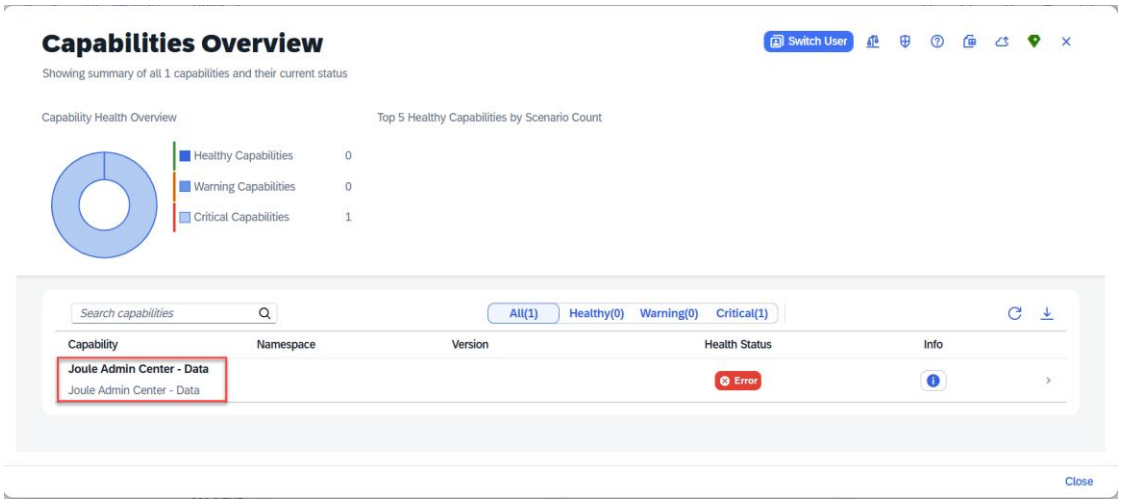


Figure: Capabilities Overview after import — Joule Admin Center – Data capability

The image below shows the scenarios loaded from the Joule Admin Center export.

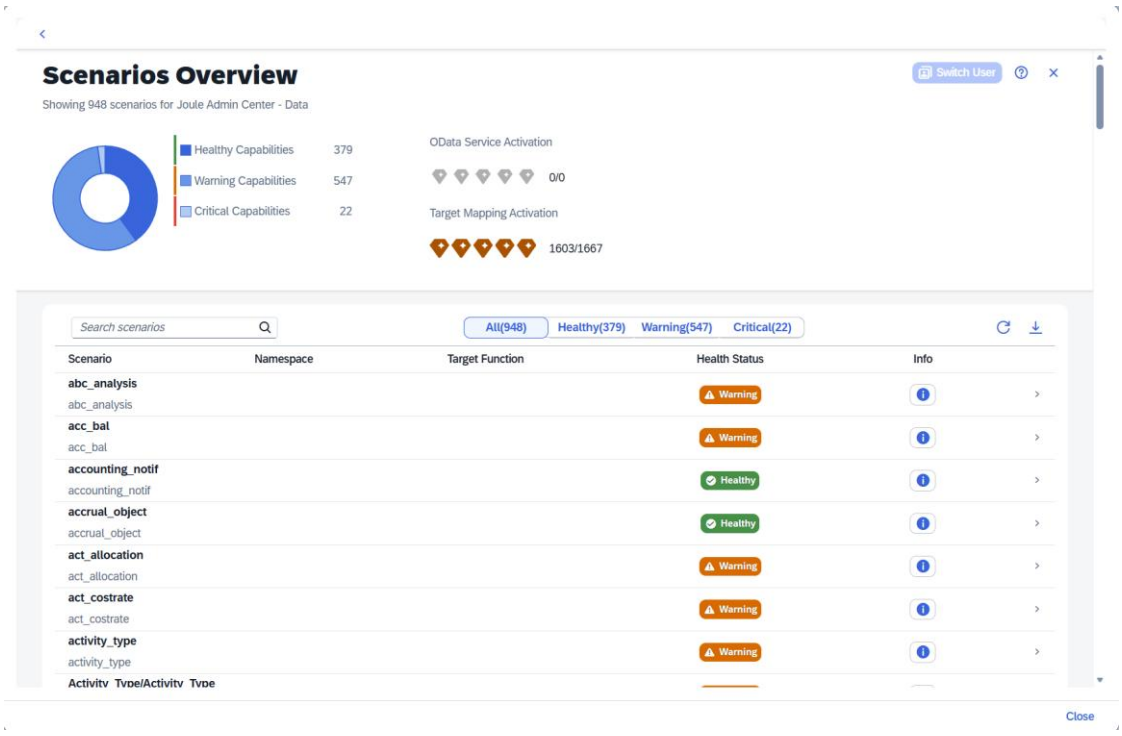


Figure: Scenarios loaded from the Joule Admin Center export

Note: This feature imports visibility condition data only. The visibility condition determines whether a Joule scenario is recognized and triggered for a given query. It does not replace or supplement the capabilities reference data from SAP Note 3523238, which must be validated separately for activation and authorization analysis.

6. Step-by-Step Workflows

6.1 Why is a Joule use case not working for the logged on user?

- **Step 1**: Open Readiness Inspection and Guidance Tool for Joule Scenarios from the FLP Shell bar. The Capabilities Overview shows all capabilities on the customer system.
- **Step 2**: Find the capability showing Warning or Critical. Click its row to open the Scenarios Overview.
- **Step 3**: Find the scenario (use case) showing the issue. Click it to open the Individual Scenario Detail.
- **Step 4**: Check the summary cards: is the issue readiness/activation (at least one artefact inactive) or access/authorization (all active but user lacks a role)?
- **Step 5**: For an activation issue: click Resolve Errors and follow the step-by-step activation guide (STC01 / STC02 instructions).
- **Step 6**: For an authorization issue: click Analyse Missing IBN Targets or Analyze Missing OData Auth. Work through the 3-column analyzer, save your role selections.
- **Step 7**: Click Resolve Errors to get the full guide with your saved role recommendations. Copy or email it to the person who assigns roles.

6.2 Analyze readiness and access for a specific context user

- **Step 1**: Press Switch User in the FLP Shell bar, search for the user on the customer system, and select them.
- **Step 2**: A yellow banner confirms you are now viewing as the selected context user.
- **Step 3**: Navigate through Capabilities, Scenarios, and Scenario Detail as normal — all data now reflects that context user's readiness and access.
- **Step 4**: Use Analyse Missing IBN Targets or APIA to identify missing roles for that context user.
- **Step 5**: Press Cancel Simulation to return to the logged on user context.

6.3 Update the capability reference data from SAP Note 3523238

- **Step 1**: Download the latest version of the Excel file "Business Roles and API for Transactional Capabilities.xlsx" from SAP Note 3523238.
- **Step 2**: Press the Excel icon (DB Sync) button in the Capabilities Overview header in the FLP Shell bar.
- **Step 3**: Select Upload mode, pick the downloaded file, and press Import.
- **Step 4**: The Import History table confirms the result. Press View Logs for detailed validation.

6.4 Analyze authorization coverage for a set of business roles

- **Step 1**: Open the Custom Role Analyzer from the Capabilities Overview screen header — press the shield icon button (third button in the header).
- **Step 2**: Add the business roles to analyse. Type a role name and use type-ahead to select it, or upload an Excel file with role names as column headers. Both methods can be combined into a single unique role list.
- **Step 3**: Select the analysis mode. For small to medium role sets choose Standard. For large role sets choose Background (requires SAP S/4HANA 2025 with bgPF and bgRFC configured and BGMC_DEFAULT active).
- **Step 4**: Press Analyze. Standard mode runs immediately in a dialog. Background mode schedules a background job — the result appears in the Analysis History table once complete.
- **Step 5**: Open the analysis results from the Analysis History table. Review the N × M coverage matrix showing whether each role covers each scenario: Covered, Partially covered, or Not Covered.
- **Step 6**: Click any cell in the matrix to see the exact missing requirements and rating for that role and scenario combination.

7. Quick Reference — All Buttons and Actions

Button	Where to find it	What it does
Switch User	Capabilities Overview screen	Switch the analysis context to any other user on the customer system.
Cancel Simulation	Capabilities Overview screen	Return to the logged on user context.
Prompt Library	Capabilities and Scenario screen headers	Open the Joule Prompt Library.
Capability Data Sync	Capabilities Overview screen	Open the ***Capability Data Sync*** screen.
Joule Plugin Status indicator	Capabilities Overview screen	Green = Joule plugin available and initialized. Red = not detected in the FLP.
Inspect icon (on OData service row)	Scenario Detail — OData services section	Launch APIA for that specific OData service.

Inspect icon (on SAP Fiori App row)	Scenario Detail — IBN Targets section	Launch TMA for that specific SAP Fiori App (IBN Target).
Resolve Errors	Scenario Detail screen header	Open the Resolve Errors screen.
Analyze Missing IBN Targets	Scenario Detail screen header	Launch TMA for all unauthorized SAP Fiori Apps (IBN Targets) at once.
Analyze Missing OData Auth	Scenario Detail screen header	Launch APIA for all unauthorized OData services at once.
Refresh	All detail screens	Reload all data for the current screen.
Export / Download	All detail screens	Download current screen data to Excel.
Save Selection	TMA and APIA screens	Save the current TM / API + Business Role + Composite Role combination.
Download (TMA / APIA)	TMA and APIA screens	Export all saved role selections to Excel.
Analyze Roles (CRA)	Custom Role Analyzer — input screen	Start the CRA analysis for the selected roles and mode.
Re-run (CRA)	Custom Role Analyzer — Analysis History table	Re-run a previously executed CRA analysis with the same role inputs.
Filter Results (CRA)	Custom Role Analyzer — analysis results screen	Filter the coverage matrix by Role, Capability, Scenario, or Coverage status.
Import Admin Center Data	Capabilities Overview screens	Open the Import Joule Admin Center Data screen to upload scenario visibility data.